

POLARIS DC-1 — MECHANICAL COOLING AND HEAT RECOVERY

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Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland
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Synthetic demonstration data. All parties, ratings and figures are fictional.

1. Scope

- 1.1 This specification covers the design, supply, installation, testing and commissioning of the mechanical cooling system serving the four data halls of Polaris DC-1, comprising indirect adiabatic free-cooling air handling units (CRAH/iAHU), chilled-water backup plant, dry coolers, pipework, controls and the waste-heat recovery interface to the district-heating boundary connection.
- 1.2 The Subcontractor shall design, procure, install, test and commission the complete mechanical cooling system in accordance with the Basis of Design (PDC-BOD-001 §6–7) and the Employer's Requirements (PDC-ER-001 §4), and shall carry design responsibility for the means of compliance except where a requirement of this specification is stated as prescriptive.
- 1.3 Excluded from this specification: structural steelwork and builder's work openings (refer PDC-SPEC-01/02), electrical supply to mechanical plant (refer PDC-SPEC-06), and the BMS field network backbone (refer PDC-SPEC-07), save that the mechanical control philosophy in Section 5 shall be read together with PDC-SPEC-07.
- 1.4 The system shall serve four independently commissionable 2.5 MW data halls sharing a common cooling spine and a shared backup-chiller zone, consistent with the sectional completion strategy of PDC-BOD-001 §2 and §9.

2. Standards

2.1 The Subcontractor shall comply with, at minimum, the following:

Reference	Application
National Building Code of Finland	Building services, ventilation and structural fire chapters
Land Use and Building Act (132/1999)	Statutory design basis
Rescue Act (379/2011)	Fire and rescue service access, smoke control coordination
SFS 6000	Low-voltage installations — control panels and mechanical plant supplies
SFS 6001	High-voltage installations — interface coordination only, no HV equipment in this specification
EN 50600 series (all parts)	Data centre facility, power and environmental control design reference
EN 50600-4-2	PUE calculation methodology — basis for the guarantee in Section 5.6
EN 13779 / EN 16798	Ventilation for non-residential buildings, indoor environmental input parameters
ASHRAE TC9.9 (2021 thermal guidelines)	Reference only, for containment and environmental envelope cross-check

IEC 60204-1	Safety of machinery — electrical equipment of machines, mechanical plant control panels
PED 2014/68/EU (national implementation)	Pressure equipment — chillers, pipework above threshold
Government Decree 331/2016 (F-gas)	Refrigerant handling, leak testing and logging

2.2 Where a cited standard has a Finnish national annex or national implementing regulation, the national version governs. Where two standards conflict, the more onerous requirement governs, per PDC-BOD-001 §1.2.

3. Equipment schedules

3.1 Design external conditions

Condition	Value	Basis
Summer design dry bulb	+29 °C	Vantaa, per PDC-ER-001 §4.1
Summer design wet bulb	+19 °C	Vantaa, per PDC-ER-001 §4.1
Winter design dry bulb	-29 °C	Vantaa, per PDC-ER-001 §4.1
White space supply air at rack inlet	24 °C ± 1.5 K	PDC-ER-001 §4.1, PDC-TA-001 §5
White space relative humidity	20–70% RH, non-condensing	PDC-ER-001 §4.1
Rack density	8 kW average / 15 kW peak per rack	PDC-ER-001 §3.1, PDC-TA-001 §5

3.2 Free-cooling hours basis

The design shall demonstrate not less than **8,300 free-cooling hours per year** (approximately 95% of 8,760 annual hours), calculated as the number of hours per year during which the indirect adiabatic system, with or without adiabatic spray assist, can meet the 24 °C ± 1.5 K rack-inlet requirement without mechanical (chiller) assistance, using a TMY3-equivalent Vantaa weather file or better. The Subcontractor shall submit the free-cooling hour calculation with the S2 detailed design submission, referencing PDC-BOD-001 §6.1, which allocates approximately 460 hours per year to adiabatic and mechanical assistance during the warmest summer conditions.

3.3 CRAH / indirect adiabatic air handling units

Item	Requirement
Quantity	28 units total (7 per hall)
Configuration	6 duty + 1 standby per hall (N+1), concurrently maintainable
Type	Indirect adiabatic air handling unit with adiabatic spray-assist stage and chilled-water backup coil
Air management	Overhead cold-aisle containment supply, no raised floor
Per-unit airflow	Sized to the hall's 2.5 MW IT load at 24 °C ± 1.5 K supply, ΔT per manufacturer's selection, confirmed at S2
Fan type	EC plenum fans, variable speed, N+1 fan redundancy within each unit
Filtration	Minimum ISO ePM1 55% (or equivalent) supply-side filtration, coordinated with PDC-TA-001 §5 air quality requirement

3.4 Chilled-water backup plant

Item	Requirement
Chillers	4 units, N+1 (3 duty + 1 standby), air-cooled, located in the shared backup-chiller zone
Arrangement	Parallel offtake from the cooling yard's shared backup-chiller zone, not in series with any hall's primary cooling path
Dry coolers	10 units, arranged across the cooling yard, serving the chiller condenser loop and the heat recovery loop

Pipework	Pre-insulated, welded, factory or site pressure-tested to 1.5× design pressure
Glycol	Concentration set to protect against the -29 °C winter design condition in any external or unheated run
Pumps	Duty/standby pairs at each hall riser and at the chiller plant room, variable speed

3.5 Waste-heat recovery plant

Item	Requirement
Thermal capacity	Not less than 4 MW thermal
Flow temperature	Not less than 70 °C at the boundary connection chamber
Availability trigger	Whenever site IT load exceeds 40% of the 10 MW design total
Arrangement	Parallel offtake from the backup-chiller zone; isolation of the heat-recovery plant shall not remove cooling capacity from an operating hall
Interface	Flanged connection at the site boundary chamber; offtaker's network excluded — refer PDC-DH-001
Seasonal note	Full-capacity testing is only achievable during the Finnish heating season (approximately October–April); the Subcontractor shall programme commissioning accordingly

4. Installation and workmanship

4.1 All pipework shall be installed within the shared MEP spine risers and coordinated within the 2.4 m overhead service zone defined in PDC-BOD-001 §3.2, above the 4.2 m minimum clear white-space height. Welded joints shall be used throughout on chilled-water and glycol circuits above 50 mm nominal bore; mechanical joints are permitted below that size subject to Engineer approval.

4.2 All pipework shall be pressure-tested at 1.5× design pressure for not less than 4 hours with no measurable pressure drop, witnessed and recorded before insulation is applied.

4.3 CRAH units shall be installed and commissioned hall-by-hall so that each hall's cooling capacity is complete and independently testable at that hall's Section handover, without requiring access to an operating hall's plant room. No penetration between a hall's mechanical plant room and the shared MEP spine shall be made without fire-stopping to EI 120, matching the compartmentation rating of PDC-BOD-001 §3.4.

4.4 Adiabatic make-up water pipework and controls shall be metered per hall and site-wide, feeding the DCIM per Section 5.5, and shall be designed to keep site-wide make-up water within the 12,000 m³/year cap set out in PDC-BOD-001 §6.2.

4.5 All rotating equipment shall be installed on vibration isolation mounts; measured vibration at rack mounting level shall not exceed 0.5 g (0–500 Hz) per PDC-TA-001 §5. Refrigerant circuits shall be leak-tested and logged in accordance with Government Decree 331/2016 prior to charging.

5. Control and sequences of operation

5.1 Modes of operation

Mode	Trigger	Description
Free cooling	Outside air wet bulb allows 24 °C ± 1.5 K supply without assist	Direct indirect heat exchange, fans only
Adiabatic-assisted free	Outside air alone insufficient	Adiabatic spray assist depresses effective entering air

cooling	temperature	
Mechanical assist / chilled-water backup	Neither mode alone meets setpoint	Chilled-water backup coil engaged, chillers running
Failure mode — loss of a chiller	Chiller fault	Remaining N+1 chillers absorb load with no loss of white space conditions, per PDC-TA-001 §4

5.2 White space setpoints

Parameter	Setpoint	Alarm band
Rack inlet supply temperature	24 °C	Alarm at ± 1.5 K sustained > 5 min
Relative humidity	45% RH midpoint	Alarm outside 20–70% RH
Differential pressure, containment	Per manufacturer's selection	Alarm on loss of positive containment pressure
CRAH fan failure	—	Immediate alarm, auto-rotation to standby unit

5.3 Alarm schedule (extract)

Alarm	Priority	Action
Rack inlet temperature out of band	Critical	BMS escalation, DCIM log, NOC notification
Adiabatic make-up water flow high	Warning	Investigate leak, check against 12,000 m ³ /year cap
Chiller fault	Critical	Auto-rotate to standby chiller, DCIM log
Heat-recovery flow temperature below 70 °C	Warning	Isolate heat-recovery offtake, notify district-heating interface
Glycol concentration low (freeze risk)	Critical	Isolate affected run, notify FM team

5.4 Free-cooling / mechanical-assist changeover

The BMS shall automatically select the lowest-energy mode capable of meeting the white space setpoint, in the order: free cooling, adiabatic-assisted free cooling, mechanical assist. Changeover shall be bumpless, with no rack-inlet excursion beyond the alarm band during transition.

5.5 Metering and DCIM integration

All chilled-water flow/return temperatures, adiabatic make-up water flow, CRAH fan status and heat-recovery flow temperature/flow rate shall be reported to the DCIM via the BMS open protocol (BACnet/IP, refer PDC-SPEC-07), with historical trending retained for not less than 24 months per PDC-ER-001 §6.

5.6 PUE and WUE contribution

The mechanical system's contribution to the annualised PUE guarantee (≤ 1.25 , calculated per EN 50600-4-2) and to the WUE guarantee (≤ 0.25 L/kWh) shall be demonstrated by the metering and reporting strategy above, referencing the PUE build-up in PDC-BOD-001 §7.

6. Testing and commissioning

Commissioning follows the five-level structure of PDC-ER-001 §8, applied per Section:

Level	Activity	Mechanical scope
L1	Factory acceptance testing	Chillers, CRAH units — witnessed factory performance test against schedule ratings
L2	Site delivery and static inspection	Visual inspection, damage check, nameplate verification against schedule
L3	Individual component start-up and functional test	Each CRAH unit, chiller, pump and dry cooler started and functionally tested per hall

L4	Integrated systems testing, including black-building and failure-mode tests	Loss-of-chiller test per PDC-TA-001 §4 with no loss of white space conditions; free-cooling to mechanical-assist changeover test
L5	Integrated load-bank testing to full Section load for 24 hours continuous	White space environmental conditions logged throughout; Section 1 tested to the full 10 MW shared cooling spine capacity per PDC-BOD-001 §9

Tests after Completion comprise the 12-month annualised PUE and WUE demonstration using metered data, per PDC-ER-001 §4.3 and §8.

7. Submittals

7.1 The Subcontractor shall submit, at the design stages defined in PDC-ER-001 §10:

- CRAH, chiller and dry cooler equipment schedules with full performance curves;
- free-cooling hours calculation per Section 3.2;
- pipework layout and riser schematics coordinated with the 2.4 m service zone;
- control philosophy and cause-and-effect narrative, cross-referenced to PDC-SPEC-07;
- water balance calculation demonstrating compliance with the 12,000 m³/year make-up water cap;
- commissioning test scripts for L1–L5, including the loss-of-chiller failure-mode test script.

8. Spares, training and warranties

8.1 The Subcontractor shall supply a spares schedule covering not less than two years of routine maintenance consumables (filters, belts, adiabatic media) and one critical spare of each rotating assembly type (fan, pump head) per hall.

8.2 The Subcontractor shall provide not less than 5 days of operator training per Section, covering normal operation, failure-mode response and DCIM interpretation, with training records forming part of the S4 handover documentation.

8.3 Equipment warranties shall run for not less than 24 months from the relevant Section's Service Commencement date, or 30 months from delivery, whichever is later, and shall be assignable to the Employer's facilities management provider.

8.4 The Subcontractor shall maintain critical-plant response cover (chillers, CRAH units) for the defects period, with a maximum 4-hour attendance time for a Critical alarm per Section 5.3.

End of PDC-SPEC-05 Rev A.